

Matteo Leonelli

Cybersecurity Researcher

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Summary

Ph.D. candidate at CISPA specializing in coverage-guided fuzzing, vulnerability research, and binary analysis, with peer-reviewed publications at top security venues (NDSS, EuroS&P) and industry research experience at Amazon Web Services (AWS). Discovered and disclosed 4+ CVEs across hardware (Intel graphics), multimedia (FFmpeg), and widely-deployed WordPress plugins. Organizing committee member for the SBFT tool competition (ICSE 2025) and mentor to Bachelor's and undergraduate research students.

Experience

CISPA Helmholtz Center for Information Security, Saarland University

Saarbrücken, Germany

PH.D. CANDIDATE, SYSTEM SECURITY

Mar. 2024 – Present

- EuroS&P'26: **THEMIS** – context-aware grey-box fuzzer for WordPress plugins; discovered multiple authenticated/unauthenticated vulnerabilities in widely-deployed plugins
- NDSS'25: **TwinFuzz** – differential testing framework for video hardware acceleration stacks, leading to Intel CVE-2024-23919
- SBFT'25: co-organized the Fuzzing Track Tool Competition at ICSE 2025
- Discovered and disclosed 4 CVEs across Intel graphics software and WordPress plugins (CVE-2024-23919, CVE-2025-9219, CVE-2025-5701, CVE-2023-7306)

RESEARCH ASSISTANT

Apr. 2022 – Mar. 2024

- Collaborated with Ph.D. researchers on automated root-cause analysis of fuzzer-discovered bugs, localizing vulnerabilities to specific commits and code versions across software histories
- Researched LLM-driven automatic fuzzing-harness generation to reduce manual effort in setting up fuzzing targets across different frameworks

MASTER'S THESIS

Mar. 2023 – Jan. 2024

- Research on software and hardware fuzzing in collaboration with Intel

Amazon Web Services (AWS)

Paris, France

APPLIED SCIENTIST INTERN – RESEARCH INTERNSHIP

Sep. 2025 – Nov. 2025

- Designed and implemented automated fuzz-testing pipelines to evaluate and ensure data integrity in Amazon S3
- Applied grey-box fuzzing techniques to cloud storage systems, identifying edge-case failure modes in large-scale distributed infrastructure

Fondazione Bruno Kessler

Trento, Italy

BACHELOR THESIS

May 2021 – Sep. 2021

- Research resulting in the CODASPY'21 paper: *Secure Pull Printing with QR Codes and National eID Cards: A Software-Oriented Design and an Open-Source Implementation*

JUNIOR RESEARCHER

Feb. 2019 – Dec. 2020

- Designed authentication protocols using the Italian CIE 3.0 smart card standard

SOFTWARE DEVELOPER INTERN, SECURITY & TRUST LAB

Nov. 2018 – Jan. 2019

- Mobile developer internship focused on secure application development

JAVA DEVELOPER INTERN, SMART COMMUNITY LAB

Jun. 2017 – Jul. 2017

- Java developer involved in data analysis and software development

Matrix Tech

Trento, Italy

FULL STACK DEVELOPER

Dec. 2017 – Dec. 2020

- Developed a full-stack web service for access control and real-time occupancy monitoring, integrated with proprietary embedded hardware

OGP

Trento, Italy

DEVELOPER

2018 – 2019

- Developed customer-facing websites and integrated new features

Education

CISPA Helmholtz Center for Information Security

Saarbrücken, Germany

PH.D. IN CYBERSECURITY

Mar. 2024 – Present

Saarland University

Saarbrücken, Germany

M.ENG. IN CYBERSECURITY

Oct. 2021 – Feb. 2024

University of Trento

Trento, Italy

B.SC. IN COMPUTER SCIENCE

Sep. 2018 – Sep. 2021

Technological Institute "G. Marconi"

Rovereto, Italy

DIPLOMA IN COMPUTER SCIENCE AND TELECOMMUNICATIONS

Sep. 2013 – Jul. 2018

Skills

Research Interests

Coverage-Guided Fuzzing, Differential Testing, Binary Analysis, Web Application Security, Hardware Security Testing
Programming

C, C++, Python, Bash, Rust, PHP

Languages

Italian (Native), English (C1), German (B1), Spanish (B1)

Publications

THEMIS: Context-Aware Grey-box Fuzzing for WordPress Plugins

Matteo Leonelli, David Jannis Dewes, Thorsten Holz

IEEE EuroS&P
2026

SBFT Tool Competition 2025 – Fuzzing Track

Addison Crump, Sahil Sihag, Matteo Leonelli

SBFT
2025

TwinFuzz: Differential Testing of Video Hardware Acceleration Stacks

Matteo Leonelli, Addison Crump, Meng Wang, Florian Bauckholt, Keno Hassler, Ali Abbasi, Thorsten Holz

NDSS
2025

Secure Pull Printing with QR Codes and National eID Cards: A Software-Oriented Design and an Open-Source Implementation

Matteo Leonelli, Umberto Morelli, Giada Sciarretta, Silvio Ranise

ACM CODASPY
2021

Pull Printing with National eID Cards: An Open-Source and Software-Oriented Implementation

Matteo Leonelli, Umberto Morelli, Silvio Ranise, Giada Sciarretta

ITASEC
2020

Vulnerabilities & Achievements

2025 **CVE-2025-9219**

Post SMTP (WordPress plugin) – missing authorization to authenticated subscriber

2025 **CVE-2025-5701**

HyperComments (WordPress plugin) – unauthenticated arbitrary options update

2024 **CVE-2024-23919**

Improper buffer restrictions in Intel Graphics software (Intel Security Acknowledgements)

2024 **Reported, FFmpeg**

Free() attempted on address that was never malloc'ed

2023 **CVE-2023-7306**

Frontend File Manager (WordPress plugin) – unauthenticated arbitrary post deletion

2023 **Reported, FFmpeg**

Global buffer overflow

Teaching & Supervision

2025–26 **Research Assistant Mentor**

CISPA Helmholtz Center for Information Security

2025 **Bachelor's Thesis Supervisor**

CISPA Helmholtz Center for Information Security

2024 **Teaching Assistant, System Security**

CISPA Helmholtz Center for Information Security

2023 **Teaching Assistant, Security**

University of Saarland

2023 **Teaching Assistant, Human Computer Interaction**

University of Saarland

2021 **Teaching Assistant, Programming 1**

University of Trento

Talks & Summer Schools

2026 **Attended IEEE S&P** *San Francisco, USA*

2026 **Attended Fuzzing and Software Security Summer School** *Singapore*

2026 **Invited Contributor, CISPA Blog Article: Themis** *CISPA*

2025 **Attended and Presented at NDSS** *San Diego, USA*

2025 **Organized SBFT Workshop** *ICSE 2025*

2025 **Invited Guest, CISPA Podcast** *Black Box Testing with Matteo Leonelli*

2024 **Attended and Presented at Intel Workshop: Scalable Assurance** *Portland, USA*

2024 **Attended EPFL Summer School in System Security** *Switzerland*

2023 **Attended CISPA Summer School in System Security** *Saarland, Germany*

2022 **Attended CISPA Summer School in Trustworthy Artificial Intelligence** *Saarland, Germany*

2021 **Attended University of Trento School of Innovation** *Trento, Italy*

2019 **Organized ProM Camp** *Trento, Italy*

2018 **Attended ProM Camp** *Trento, Italy*